

Noma: An Open System One Decision Model from a Sliced Decoder and Trained Heads

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OPEN RESOURCES [Code](#) | [Model weights](#) | [Project page](#) | `pip install blackdrome-noma`
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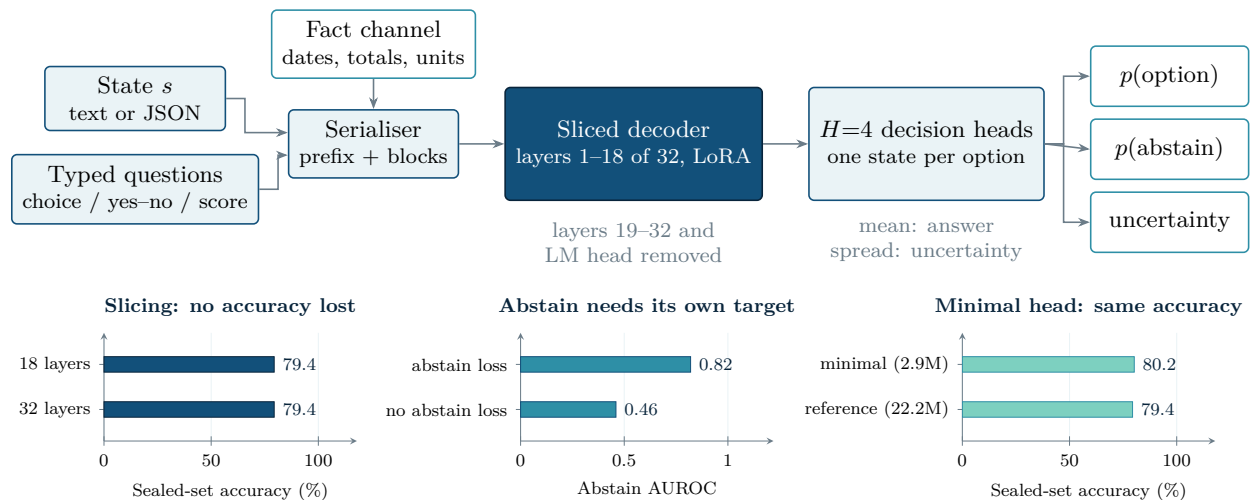


Figure 1 | Noma and what each part buys. Top: a state and typed questions are serialised with computed fact lines, read once by the first 18 layers of a pretrained decoder, and scored by trained heads that read one hidden state per option; nothing is generated. Bottom: three results from the 6,000-item ablations. Slicing to 18 of 32 layers loses no measured accuracy; the abstain output is informative only when it is supervised; a single small head scores within two points of the four-head reference (difference +0.8, interval $[-1.3, +2.8]$). AUROC is the area under the ROC curve.

Abstract

System One decision models answer typed questions about a state in a single forward pass, returning a probability for every option and no generated text. The class was introduced with Jev, a hosted model. We present NOMA, an open-weight model in this class, and a measured account of its design: a pretrained 4B decoder sliced to its first 18 of 32 layers, trained decision heads that read one hidden state per option, a separately supervised abstain head, and a fast serving path. On single-pass decisions NOMA scores 82.6% on a sealed, human-reviewed set that we also used to compare development runs (81.3% for a fresh re-training of the same design), with mean confidence about two points above accuracy. On two intent tasks absent from its training data it reaches 83.6% (150 options) and 70.1% (77 options) with no adaptation, and it answers in 15 ms at the median on one H100 with zero output tokens. Controlled ablations (20 runs at a 6,000-item recipe, paired per item) show which parts carry this. Slicing loses no

measured accuracy (79.4% against 79.4% over three seeds each; 95% interval on the difference $[-2.6, +2.7]$). The abstain output is informative only when it is supervised (area under the ROC curve 0.82 against 0.46, one seed, 15 items). The option head can be small: one pointwise head about an eighth the size of the four-head listwise reference scores 80.2% against 79.4% (difference $+0.8 [-1.3, +2.8]$), and with all the training data the two designs remain within noise of each other. A single pass has limits: on 56 held-out items that need chained computation, every configuration with at least 18 layers scores between 44.6% and 62.5% (chance is 33.5%), the released model’s confidence is 23 points above its accuracy, and 9.7% of multiple-choice answers change when the options are reversed. Code and weights are open.

1. Introduction

An agent that plans, calls tools and checks its own work makes many small decisions between the steps that need generation: which queue a ticket belongs to, whether a tool call succeeded, whether the next action is permitted by a policy, whether the task is finished. Each is a question with a fixed set of answers. Answering it with a general language model costs a decoding loop and returns text that must be parsed, with a confidence that is at best a by-product.

System One decision models were introduced for this job. Jev [2], the first of the class, takes a program state and typed questions and returns structured values with probabilities, without generating a string. It is a hosted model; its announcement describes a new architecture trained with reinforcement learning for calibrated decisions and reports end-to-end latencies of 70 to 500 ms. An open-weight model with the same interface has followed, and an independent evaluation of the two has recently appeared [24]. What is still missing is an open, documented design: published systems report end results, not which choices produced them.

We present NOMA, an open-weight System One decision model that accepts the same requests as Jev, and we ask one question of its design: *which parts of a decision model buy accuracy or calibration, and which can be left out?* NOMA combines four parts (Figure 1, Section 3): a pretrained 4B decoder *sliced* to its first 18 of 32 layers; *trained decision heads* that read one hidden state per option, so that a decision is one forward pass with no output tokens and the options are free at request time; a *dedicated abstain head* with its own supervision and calibration; and a *fast serving path*.

On single-pass decisions, the released model is accurate, calibrated and fast (Section 5). It scores 82.6% on a sealed, human-reviewed set of 386 decisions that was also used to compare development runs (a fresh re-training of the same design scores 81.3%), with a mean confidence of 84.7%; when it answers only the more confident half of them, it is right 96.9% of the time. On two intent tasks that are not among its training sources, it reaches 83.6% with 150 options and 70.1% with 77, without any adaptation. It answers in 15 ms at the median on one H100, one request at a time.

Ablations show which parts of the design carry this (Section 6). Across 20 controlled runs with per-item paired comparisons, slicing loses no measured accuracy: 18 layers and all 32 score 79.4% and 79.4% over three seeds each, with a 95% interval of $[-2.6, +2.7]$ points on the difference, and accuracy plateaus near layer 18. The abstain output is informative only when it is supervised. The option head can be small: a single pointwise head about one eighth the size of the four-head listwise reference scores 80.2% against 79.4% (difference $+0.8 [-1.3, +2.8]$), and with all the training data the two designs remain within noise of each other.

A single pass has limits (Section 7). On 56 held-out items that need chained computation, every configuration with at least 18 layers scores between 44.6% and 62.5%, against 33.5% for chance, and the model’s confidence there is 23 points above its accuracy. Depth beyond the plateau

and about 3,800 targeted synthetic training items did not close the gap. One multiple-choice answer in ten changes when the options are listed in reverse.

Contributions.

- An open System One decision model, with weights, code and serving path, compatible with the Jev request format.
- A documented architecture and an ablation of its parts: depth, head type, ensembling, bootstrap resampling, a computed fact channel, abstain supervision and two loss terms, with seed-aware intervals and paired per-item statistics.
- A held-out multi-step split that no data generator or training run saw, and a negative result on it.

The released model was trained before the ablations were run, so it carries the full design, including parts the ablations later showed to be unnecessary at this scale. We report the model as released and the ablations as measured.

2. Problem: What a Decision Model Must Do

A System One decision model receives a *state* s (text or structured data) and one or more typed *questions*. A question is a *choice* among K options supplied with the request, a *yes or no*, or a *score* on an ordered scale. For each question the model returns a probability for every option. We take the task and the request format from Jev [2]; Figure 2 shows a request and its response. Four requirements follow from how such a model is used, and they organise the rest of the paper.

Requirement 1: one fast forward pass. The decision sits in the inner loop of an agent or a pipeline, so its latency and cost are paid at every step. The model should read the state once, answer all questions about it, and generate no tokens.

Requirement 2: probabilities that can be acted on. The caller automates what the model is sure of and escalates the rest. That only works if stated confidence tracks accuracy, so that a threshold chosen on one day’s traffic holds on the next.

Requirement 3: a way to say “none of these”. Callers offer the options they thought of. When none is supported by the state, the model should say so through a signal that is separate from the option probabilities, which must still sum to one.

Requirement 4: open option sets. Options arrive with the request. Their number and wording are not known at training time, so the model cannot have a fixed label space.

Section 3 describes how NOMA is built to meet these. Section 5 tests the released model against them, Section 6 asks which parts of the design are needed for that, and Section 7 reports where they are not met.

3. Noma: Architecture

NOMA maps a state s and typed questions $q_1, \dots, q_m$ to, for each question, a distribution over its K options, an abstain probability and an uncertainty. Figure 3 shows the complete architecture. After a serialiser, it has the four parts named in the introduction: the sliced backbone, decision heads, an abstain output, and a serving path. Each part answers one of the requirements of Section 2: the sliced backbone and the serving path serve the single fast pass (Requirement 1), the objective and temperature scaling target usable probabilities (Requirement 2), the abstain head gives a separate “none of these” signal (Requirement 3), and reading options from marker states keeps the option set open (Requirement 4).

```

POST /v1/systemone
{"state": "Order 8841 was charged twice
    on 3 March. Customer asks
    for one charge back.",
"questions": {"team": {
    "type": "choice",
    "instructions": "Who handles this?",
    "criteria": {
        "billing": "charges, refunds",
        "identity": "login, account",
        "shipping": "delivery"}}}}

{"model": "noma",
"answers": {"team": {
    "type": "choice",
    "choice": "billing",
    "confidence": 0.93,
    "probabilities": {"billing": 0.93,
        "identity": 0.05, "shipping": 0.02},
    "noma": {"abstain": 0.02,
        "uncertainty": 0.004}}},
"usage": {"input_tokens": 212,
    "output_tokens": 0}}

```

Figure 2 | A request and its response (values illustrative). Each question has a type, instructions and options; the response gives a probability for every option, and NOMA adds an abstain probability and an uncertainty. No tokens are generated.

3.1. Input serialisation and the fact channel

A request is serialised as one *prefix* followed by one *block* per question. The prefix holds the state, followed by computed fact lines. A block holds the question’s instructions, then each option followed by a marker token, then an abstain marker, and closes with an end token. Nine special tokens mark these boundaries. They have their own small trained embedding table, so the backbone’s embedding matrix stays frozen. The prefix has a budget of 4,096 tokens; a longer state keeps its head and tail.

A single-pass model has no scratchpad, and language models are unreliable at date arithmetic and running totals. A deterministic preprocessor therefore writes short fact lines after the state: the reference time, relative dates resolved against it, elapsed hours and inclusive day counts between timestamps, running totals in document order and where they cross a threshold named in the text, and unit conversions. It reads the questions as well as the state, so that it computes the quantities they ask about. The facts are added to the input and never forced on the output, and the model is trained with them present.

3.2. Sliced backbone

The backbone is the first L layers of a pretrained decoder, adapted with LoRA (rank 16, $\alpha=32$); the remaining layers, the final normalisation and the language-model head are discarded. We use Qwen3.5-4B-Base¹, a hybrid of linear-attention and full-attention layers, with $L=18$ of 32. We chose L with a layer-wise probe on the frozen backbone (not reported here) and then tested the choice with fine-tuned comparisons under an identical recipe (Section 6.1). Truncation removes 44% of the layers from every forward pass; the larger part of the latency reduction comes from the serving path below.

3.3. Decision heads

Option scorer. Figure A1 shows the input layout and the head. For each question we gather the hidden state at each option’s marker, at the abstain marker, and at the block’s end token, giving $K+2$ vectors: K options, the abstain slot, and a question summary that has seen all of them. The released model scores them with a *listwise* head, a two-layer bidirectional transformer of width 512 over this short sequence that emits one logit per option, so every option attends to every other. Questions on an ordered scale add an ordinal embedding per level. The alternative we test is a

¹<https://huggingface.co/Qwen/Qwen3.5-4B-Base>

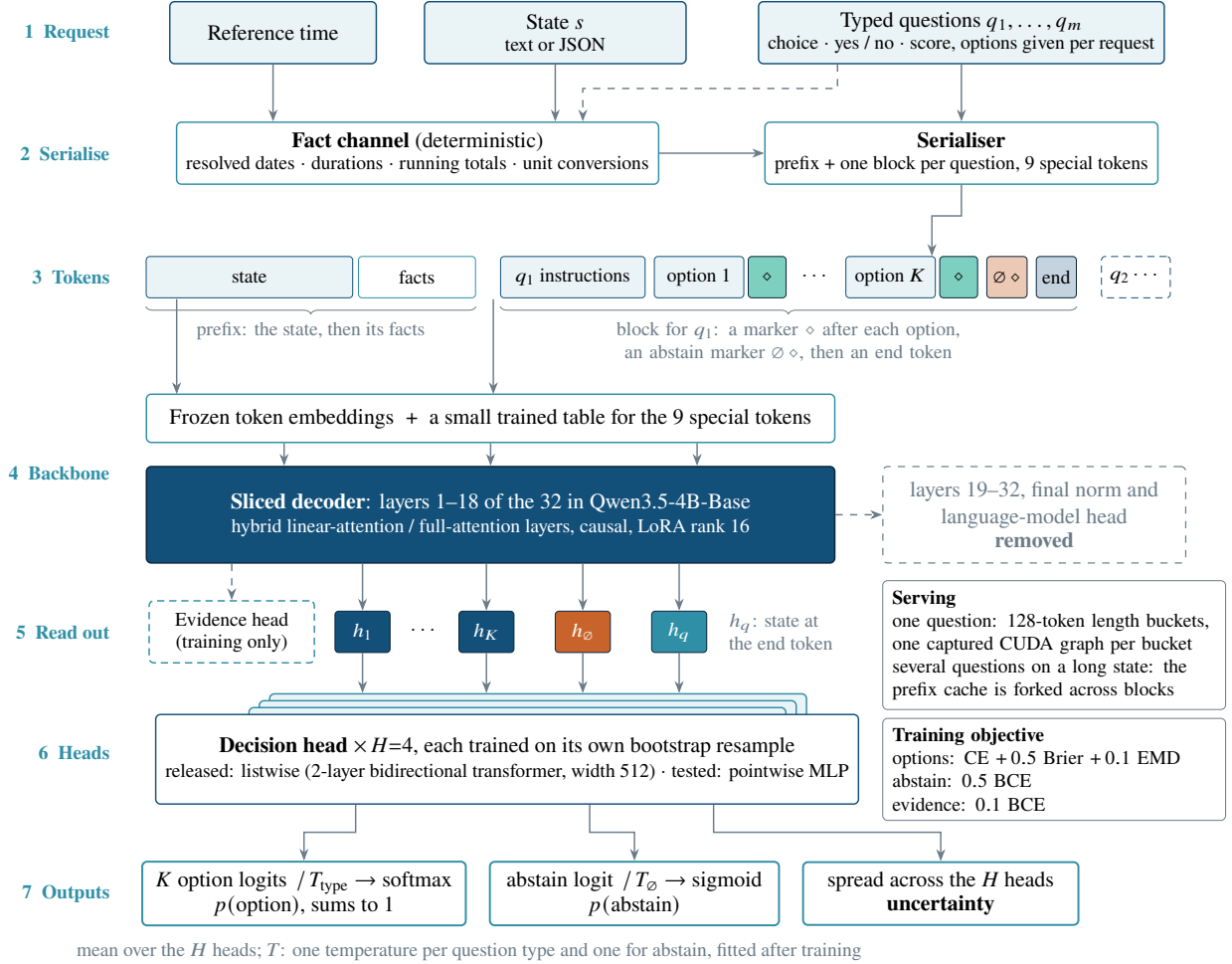


Figure 3 | The complete Noma architecture, from request to outputs. (1) A request carries a state, a reference time and typed questions with their options. (2) A deterministic fact channel computes dates, durations, totals and unit conversions from the state, the reference time and the question text, and the serialiser lays out a prefix (the state, then its facts) and one block per question. (3) In a block each option is followed by a marker token; an abstain marker and an end token close it. (4) The sequence is read once by the first 18 layers of the pretrained decoder; the remaining layers, the final normalisation and the language-model head are removed. (5) Only the hidden states at the markers and at the end token are read out; the end-token state h_q summarises the question after all its options. (6) $H=4$ decision heads score them. (7) Their mean gives a probability for every option, a separate abstain probability comes from the abstain marker, and their spread gives the uncertainty. Dashed parts are used in training only or are removed. The side boxes give the training objective and the serving path.

pointwise head, an MLP applied to each option state concatenated with the question summary, with no interaction between options.

Listwise head. **Pros:** every option sees every other, including later ones that a causal backbone has not shown it; it can in principle resolve near-duplicates. **Cons:** 5.6M parameters per head, and it adds capacity on top of a backbone that has already read the options.

Pointwise head. **Pros:** 2.9M parameters per head, each option scored independently and in parallel, no dependence on the number of options. **Cons:** it cannot compare options itself and relies on the backbone to have done so. Section 6.2 measures the difference.

Abstain as a separate output. What we call the abstain head is one extra output on each scorer: the abstain slot has its own logit, trained with a binary target that is 1 when no option is supported by the state. It is reported next to the option distribution and not inside it, so the option probabilities always sum to one and the abstain probability has its own temperature and can be thresholded separately.

Shared-backbone ensemble. We attach $H=4$ scorers to the same backbone. In the released model each is trained on its own resample of the data through an online Poisson bootstrap: every item receives a $\text{Poisson}(1)$ weight per head, fixed by a hash of the item identifier and the seed. The prediction is the mean of the heads’ distributions and the uncertainty is their disagreement, which we measure as the mutual information between the head index and the predicted option. The backbone runs once, so the ensemble adds only the cost of three more small heads.

Evidence head. A linear head over the state tokens is trained to mark the span that decides the answer, using supporting quotes from the labels as per-token targets. It is an auxiliary training signal and is not used at inference.

3.4. Objective and calibration

Targets are distributions and not single labels, so that disagreement between annotators is kept. For head h with option distribution p_h , soft target y , abstain logit $\hat{a}_h$ and abstain target a ,

$$\begin{aligned} \mathcal{L}_h = & \text{CE}(y, p_h) + 0.5 \text{Brier}(y, p_h) + 0.5 \text{BCE}(a, \hat{a}_h) \\ & + 0.1 \text{EMD}(y, p_h) \mathbf{1}[\text{ordered}], \end{aligned} \quad (1)$$

weighted per item by that head’s bootstrap count, plus 0.1 times the evidence loss. The Brier term is meant to pull probabilities towards honest values; the earth mover’s term makes an answer one level off cheaper than one three levels off. After training, one temperature per question type and one for abstain are fitted on a calibration split that is disjoint from the training items [15].

3.5. Serving

The server has two paths. *Single-question requests*, the case behind every latency figure in this paper, are padded to a multiple of 128 tokens and run through a CUDA graph captured for that length at start-up [30]; capture requires a forward pass with no host–device synchronisation. On an H100 a decision on a short state then takes about 13ms end to end. *Requests with several questions about a long state* (more than two questions and a prefix over 1,024 tokens) instead process the prefix once and expand its cache across the question blocks, which run as one batch; for a hybrid backbone the cache holds the recurrent and convolution state of the linear-attention layers as well as attention keys and values. This path matches independent runs in our consistency check, uses no captured graphs, and was not timed here. (Our first working server, before these changes, took about 2.3s per decision, a development figure that we did not log.)

4. Experimental Setup

This section describes the evaluation sets, the training runs and the statistics used in Sections 5 to 7.

Evaluation sets. (i) A *sealed set* of 401 decisions in 12 families, of which 386 have a definite answer and are scored for accuracy and 15 have no supported option and are used only to test the abstain output. The families are routing, intent, policy, severity, adequacy, relevance, tone and safety, trade-off, temporal and numeric, adversarial, ambiguous and option picking. The set is human reviewed, never used for training and never shown to a data generator. We did use it to compare development runs, so it is a development-test set and not an untouched one. (ii) The public *easy* (48 items) and *original* (72) tiers of JevBench², a benchmark for decision models. (iii) The benchmark’s public *hard* tier of 111 items, built from questions that need several chained steps, which we split in two before generating any data, stratified by family. We used 55 items as abstract structure templates for synthetic training data (the structure of the problem, never its text); we call these *exposed hard*. The other 56 were seen by no generator and no training run; this *held-out hard* set is the only multi-step set from which no training data derives, although it was scored during development. The two sets are disjoint and are always reported separately. Chance accuracy, the mean of one over the number of options, is 30.7% on the sealed set and 33.5% on held-out hard.

Controlled runs. All comparisons train on the same 6,000-item subset for one epoch with the recipe in Appendix A, changing one factor at a time. The *reference* configuration is 18 layers, listwise head, four bootstrap heads, fact channel on, and the full objective of Eq. 1. We ran three seeds for the reference, for full depth, for the pointwise head and for a minimal configuration, and one seed for every other variant: 20 runs in all. Every run stores its per-item predictions, so all comparisons are paired on items. The *released model* is the reference configuration trained on the full set of 34,408 decisions; it was chosen for release using the sealed set. Two further runs train on the full set and are reported separately in Section 6.3. Differences are computed from counts and can differ by 0.1 from the difference of two rounded figures.

Statistics. For a single accuracy we give a 95% Wilson interval. For a comparison with three seeds per arm we give the difference of mean accuracies with a 95% interval from a bootstrap that resamples both seeds and items, so that it carries seed variance. For a comparison between two single runs we give a paired bootstrap interval over items and an exact McNemar test; these carry no seed variance, and we treat them as screening results. The sets are small: an interval on one accuracy is about ± 4 points on the sealed set and ± 13 on held-out hard, and two runs of the reference configuration with different seeds are right on one and wrong on the other for 7% of sealed items and 15% of held-out hard items. When we say a component has “no detected effect” we mean the interval contains zero, and we give the interval so the reader can see what size of effect it leaves open.

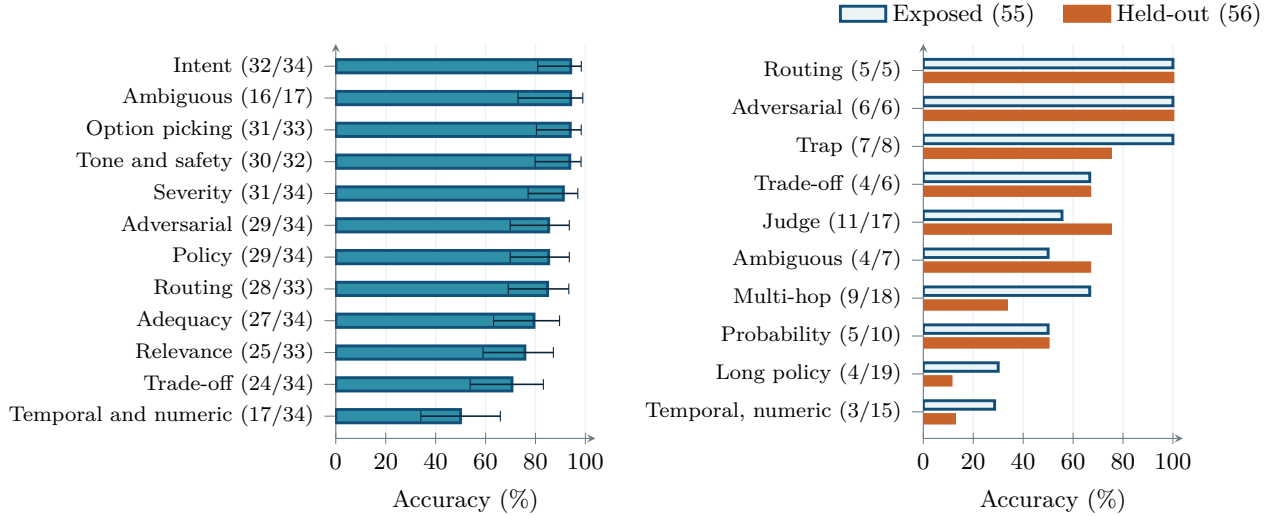
5. Results: Accurate, Calibrated and Fast on Single-Pass Decisions

We first ask whether the released model meets the four requirements of Section 2. Table 1 gives its accuracy, confidence and calibration on every evaluation set; the multi-step rows are discussed in Section 7.

²<https://github.com/fstandhartinger/jevbench>

Table 1 | The released model (18 layers, listwise head, full training set), from one per-item evaluation. Chance is the mean of one over the number of options. [†]The GPU fast path used for the published model card scores 26 of 56 on held-out hard; the two paths differ by numerical precision.

Set	n	correct	acc. $\uparrow$	95% CI	mean conf.	ECE $\downarrow$	chance
Sealed	386	319	82.6	[78.5, 86.1]	84.7	0.043	30.7
Easy	48	48	100.0	[92.6, 100.0]	98.9	0.011	28.4
Original	72	71	98.6	[92.5, 99.8]	91.7	0.071	31.1
Exposed hard	55	31	56.4	[43.3, 68.6]	71.0	0.147	33.8
Held-out hard [†]	56	27	48.2	[35.7, 61.0]	70.8	0.282	33.5



(a) Sealed set, by family, with 95% Wilson intervals. **(b) Hard tier, by family, split into exposed and held-out items.**

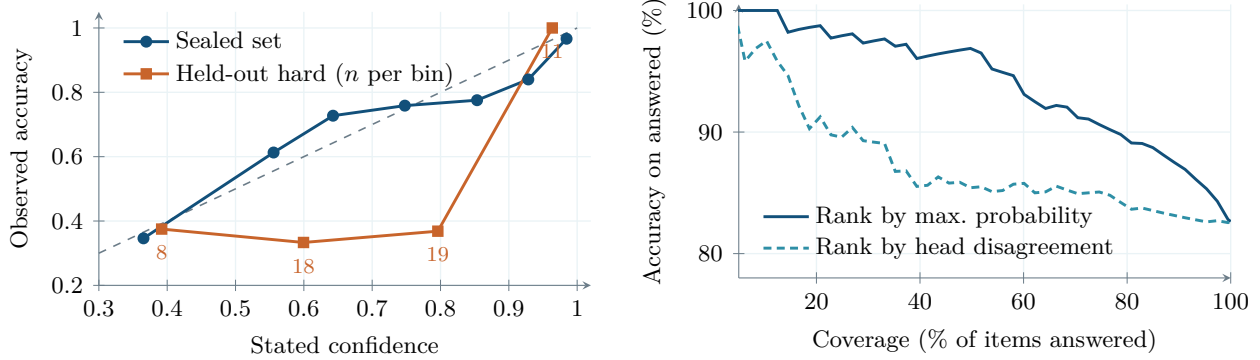
Figure 4 | Where the released model is right and wrong. On the sealed set every family but one is above 70%; the exception, temporal and numeric, is the family that needs computation. On the hard tier the same split appears: judgement families are mostly answered, calculation chains are not.

5.1. It is accurate where the answer can be read off the state

On the sealed set the released model answers 319 of 386 decisions correctly (82.6%, 95% interval [78.5, 86.1]), against a chance level of 30.7%. It answers all 48 easy-tier items and 71 of the 72 original-tier items. Figure 4 breaks this down by decision family: eleven of the twelve sealed-set families are above 70%, and the weakest, temporal and numeric, is the one that asks for computation.

5.2. Its confidence can be acted on

Calibrated where it is accurate. On the sealed set the released model’s mean confidence is 84.7% and its accuracy 82.6% (ECE 0.043). On held-out hard its mean confidence is 70.8% and its accuracy 48.2%, an overconfidence of 23 points that does not depend on binning; binned ECE there is 0.282, a figure that is biased upward on 56 items and that we give only for comparison with other work. Figure 5a shows where the gap sits: between 0.5 and 0.9 stated confidence the model is right about a third of the time. This is the expected failure of post-hoc temperature scaling under



(a) **Reliability of the released model.** Sealed-set bins have at least 22 items; the four held-out bins (confidence below 0.5, 0.5 to 0.7, 0.7 to 0.9, above 0.9) hold 8, 18, 19 and 11 items.

(b) **Selective prediction on the sealed set.**

Figure 5 | What the confidence is worth. On single-pass decisions stated confidence tracks accuracy and can be thresholded. On multi-step items the model is overconfident between 0.5 and 0.9 and reliable only above 0.9. Head disagreement ranks errors worse than the maximum probability does.

shift [35], and it means a confidence threshold below 0.9 will not catch multi-step questions. On the sealed set thresholding works: answering only the most confident half gives 96.9% accuracy, and the most confident 80% gives 89.6% (Figure 5b).

Figure A4 shows the two confidence distributions and what thresholding buys on each set.

Unseen label sets. As a check under a different kind of shift, we scored the released model on samples of two public intent-classification test sets that are not among its training sources (other intent datasets are), presenting all 77 and all 150 intents as options: Banking77 [6] (462 items) and the in-scope part of CLINC150 [22] (450 items). It reaches 70.1% [65.8, 74.1] and 83.6% [79.8, 86.7] with ECE of 0.060 and 0.070. Calibration holds when the options change and the task stays single-pass. On 150 further CLINC150 queries that belong to none of the intents, the abstain probability ranks them above in-scope queries (area under the ROC curve 0.97 [0.95, 0.98]) but stays far below 0.5, so it separates them only with a threshold tuned for that use.

Head disagreement adds little. We built the ensemble to supply an uncertainty estimate. As an error detector, disagreement between heads is weaker than the maximum probability of the mean prediction (Table A4): 0.63 against 0.82 on the sealed set for the reference. For the released model, which is not in the table, the figures are 0.59 and 0.81. Heads that share a backbone and see the same batches agree too often to be informative, with or without the bootstrap, and a single head gives the same error detection from its own maximum probability.

5.3. It answers in milliseconds

On a clean install on one H100, served over HTTP and driven one request at a time, the released model answers the 231 public benchmark items with a median of 15.1 ms end to end and a 95th percentile of 83.8 ms (Appendix C). The 95th percentile comes from the longest hard-tier states and from four requests that took 0.5 to 1 s, all in one length bucket, whose cause we have not identified. The serving path is described in Section 3.5. We did not measure inference latency at 32 layers on the fast path. In training on shared cloud GPUs, a step at 18 layers took about two thirds of the wall-clock time of a step at 32, with run-to-run timing noise of similar size. Figure A6 gives the latency by tier.

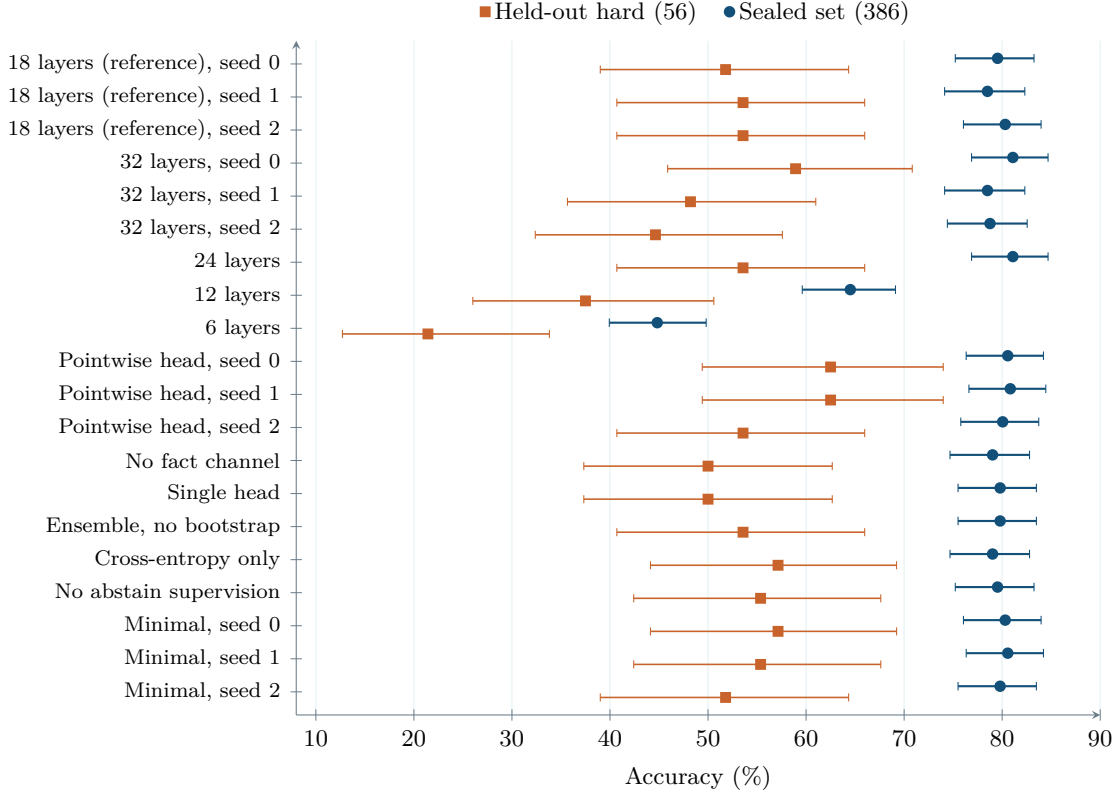


Figure 6 | Every controlled run, with 95% Wilson intervals. On the sealed set all runs with at least 18 layers fall inside each other’s intervals, whatever the head, the loss or the ensemble; only depth below 18 layers moves it. On the 56 held-out multi-step items the intervals are about ± 13 points and no configuration with at least 18 layers separates from the rest.

6. Ablations: What Each Part of the Design Buys

Figure 6 shows all 20 controlled runs at once, and is the summary of this section: at 18 layers or more, no change to the head, the loss or the ensemble moves the sealed set outside the interval of any other run.

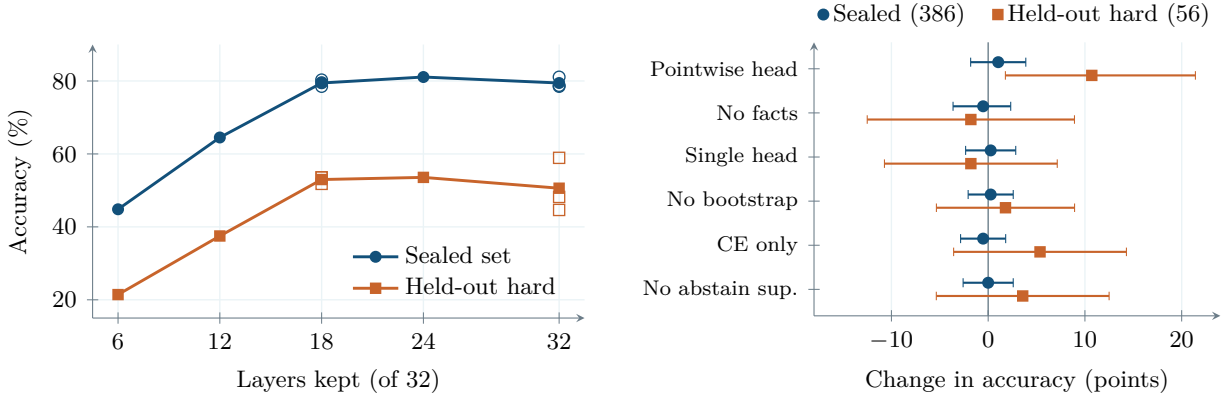
6.1. Slicing: 18 layers do the work of 32

No difference between 18 and 32 layers. Table 2 gives the comparison. On the sealed set the two depths have the same mean accuracy over three seeds, 79.4% against 79.4%. The per-seed differences are +1.6, +0.0, −1.6 points and the interval on the mean difference is $[-2.6, +2.7]$, so full depth is worth less than about three points on this set, if anything. Mean sealed-set ECE is similar, 0.033 at 18 layers and 0.031 at 32. On the two multi-step sets the point estimates favour 18 layers by 2 to 3 points, with intervals of ± 10 to ± 13 that exclude neither a real cost nor a real benefit of depth.

Where the plateau starts. Figure 7a and Table A1 add single-seed runs at 6, 12 and 24 layers. Sealed accuracy climbs from 44.8% at 6 layers to 64.5% at 12 and 79.4% at 18, then stays within two points to full depth (81.1% at 24). Held-out hard has the same shape at a lower level: 21.4%, 37.5%, 53.0%, 53.6%, 50.6%. Accuracy on multi-step items does depend on depth up to the

Table 2 | Eighteen layers against all 32, three seeds each, same data and recipe. Accuracy in %, seeds 0/1/2 and their mean. The difference is 32 minus 18; its interval resamples seeds and items.

Set	n	18 layers		32 layers		diff.	95% CI
		seeds	mean	seeds	mean		
Sealed	386	79.5 / 78.5 / 80.3	79.4	81.1 / 78.5 / 78.8	79.4	+0.0	[−2.6, +2.7]
Easy	48	100.0 / 100.0 / 100.0	100.0	100.0 / 97.9 / 100.0	99.3	−0.7	[−4.2, +0.0]
Original	72	90.3 / 98.6 / 91.7	93.5	95.8 / 88.9 / 91.7	92.1	−1.4	[−8.3, +5.1]
Exposed hard	55	54.5 / 56.4 / 47.3	52.7	56.4 / 38.2 / 54.5	49.7	−3.0	[−16.4, +9.1]
Held-out hard	56	51.8 / 53.6 / 53.6	53.0	58.9 / 48.2 / 44.6	50.6	−2.4	[−12.5, +8.9]



(a) Accuracy against depth. Lines join means; open markers are the individual seeds at 18 and 32 layers; 6, 12 and 24 layers are one seed each. Chance is 30.7% on the sealed set and 33.5% on held-out hard.

(b) Each single-seed ablation minus the reference (seed 0), with 95% paired bootstrap intervals over items (no seed variance).

Figure 7 | Depth and components. Accuracy rises until about layer 18 and is flat beyond it. No ablation moves the sealed set by more than about a point; on the 56 held-out multi-step items the intervals are wide, and the one that excludes zero (pointwise head) does not survive two further seeds (Table A3).

plateau, and stops improving in the same range as everything else. With no run between 12 and 18 layers, the sweep places the start of the plateau somewhere in that interval.

Figure A2 shows the same sweep for all five sets and for calibration error.

6.2. Head and loss: only abstain supervision shows an effect

Table 3 and Figure 7b remove one component at a time from the reference, one seed each. Across the six ablations and five sets we ran 30 paired tests without correction (Table 3 shows the 18 on the sealed set and the two hard sets), and none of them reaches $p < 0.05$; one item-level bootstrap interval excludes zero, for the pointwise head on held-out hard (McNemar $p = 0.07$), and we follow it up with more seeds below. **No ablation changes sealed-set accuracy by more than about a point**, and every sealed-set interval lies within about ± 4 points.

The pointwise head is not worse than the listwise head. This was the comparison we expected the listwise head to win. On seed 0 the pointwise head was ahead on held-out hard by +10.7 points ($p = 0.07$), which is why we ran two more seeds; a result that prompts its own replication is biased upward, so we report the replication seeds separately. On seeds 1 and 2 alone the pointwise head is ahead by +1.0 on the sealed set and +4.5 on held-out hard, level on exposed

Table 3 | Component ablations at 18 layers, seed 0, one run each. Accuracy in %; differences are ablation minus reference with paired bootstrap 95% intervals over items (no seed variance). ECE is on the sealed set, after temperature scaling.

Variant	Sealed (386)		Exposed hard (55)		Held-out hard (56)		ECE ↓
	acc. ↑	diff. [CI]	acc. ↑	diff. [CI]	acc. ↑	diff. [CI]	
Reference	79.5	–	54.5	–	51.8	–	0.029
Pointwise head	80.6	+1.0 [−1.8, +3.9]	60.0	+5.5 [−1.8, +14.5]	62.5	+10.7 [+1.8, +21.4]	0.025
No fact channel	79.0	−0.5 [−3.6, +2.3]	47.3	−7.3 [−16.4, +0.0]	50.0	−1.8 [−12.5, +8.9]	0.021
Single head	79.8	+0.3 [−2.3, +2.8]	52.7	−1.8 [−9.1, +5.5]	50.0	−1.8 [−10.7, +7.1]	0.032
Ensemble without bootstrap	79.8	+0.3 [−2.1, +2.6]	50.9	−3.6 [−12.7, +5.5]	53.6	+1.8 [−5.4, +8.9]	0.032
Cross-entropy only	79.0	−0.5 [−2.8, +1.8]	49.1	−5.5 [−12.8, +1.8]	57.1	+5.4 [−3.6, +14.3]	0.038
No abstain supervision	79.5	+0.0 [−2.6, +2.6]	56.4	+1.8 [−3.6, +9.1]	55.4	+3.6 [−5.4, +12.5]	0.057

hard (−0.9), and behind by 4.2 on the original tier. Over all three seeds (Table A3) the differences are +1.0 [−1.1, +3.2] on the sealed set and +6.5 [−3.0, +16.1] on held-out hard. Every seed-aware interval contains zero. The supported conclusion is that cross-option attention in the head brings no detected benefit, and that on the sealed set any benefit is smaller than about one point. The two heads differ in size as well as in kind (11.7M against 22.2M parameters for four heads), so this comparison does not separate the two; it shows that the smaller, simpler head is enough. A plausible reason is that the backbone has already compared the options by the time it writes each marker state, since each marker attends to the options before it.

The fact channel, the bootstrap, the ensemble and the extra loss terms: no detected effect on accuracy. Removing the fact lines changes the sealed set by −0.5 points [−3.6, +2.3] and exposed hard by −7.3 [−16.4, +0.0]; the second leaves open a real benefit of the facts on multi-step items that one seed on 55 items cannot confirm. Four heads with identical item weights are indistinguishable from four bootstrap heads, in line with Nixon et al. [32]. One head is as accurate as the mean of four. Dropping the Brier and earth mover’s terms changes sealed-set accuracy by half a point and moves sealed-set ECE from 0.029 to 0.038, just above the range of the three reference seeds (0.029, 0.036, 0.035). All ECE figures are after temperature scaling, which can absorb what these loss terms contribute; on the calibration split before scaling, ECE is 0.043 for the reference and 0.049 without the two terms.

Abstain supervision matters for the abstain output. Without the abstain loss, option accuracy is unchanged, but the abstain probability stops separating items that have no supported answer from the rest: the area under the ROC curve falls from 0.82 [0.70, 0.93] to 0.46 [0.32, 0.60] on the sealed set, and sealed-set ECE of the option distribution rises from 0.029 to 0.057, the largest calibration change in Table 3. Only 15 sealed items are of this kind, so the intervals are wide, but they do not overlap, and the direction is what one expects from removing the only loss term that trains that output.

6.3. A minimal model scores within two points of the full design

The ablations above remove one component each from one seed, and effects that are individually undetectable could add up. We therefore trained the configuration that the ablations suggest as a single model, three seeds: 18 layers, one pointwise head (so no ensemble and no bootstrap), cross-entropy without the Brier and earth mover’s terms, with abstain supervision, the evidence loss and the fact channel kept. Its head has 2.9M parameters against 22.2M for the reference. Table 4 compares it with the reference over three seeds each. On the sealed set the minimal configuration scores 80.2% against 79.4%, a difference of +0.8 points [−1.3, +2.8]; on held-out hard the difference

Table 4 | Minimal configuration (one pointwise head; cross-entropy, abstain and evidence losses) against the reference (four bootstrap listwise heads, full objective), three seeds each at 18 layers. The difference is minimal minus reference; its interval resamples seeds and items.

Set	n	Reference		Minimal		diff.	95% CI
		seeds	mean	seeds	mean		
Sealed	386	79.5 / 78.5 / 80.3	79.4	80.3 / 80.6 / 79.8	80.2	+0.8	[−1.3, +2.8]
Easy	48	100.0 / 100.0 / 100.0	100.0	100.0 / 100.0 / 100.0	100.0	+0.0	[+0.0, +0.0]
Original	72	90.3 / 98.6 / 91.7	93.5	93.1 / 95.8 / 93.1	94.0	+0.5	[−5.6, +6.5]
Exposed hard	55	54.5 / 56.4 / 47.3	52.7	52.7 / 50.9 / 47.3	50.3	−2.4	[−10.9, +5.5]
Held-out hard	56	51.8 / 53.6 / 53.6	53.0	57.1 / 55.4 / 51.8	54.8	+1.8	[−6.5, +9.5]

Table 5 | Full design against the minimal configuration when both are trained on all 34,408 training items (one seed each). The difference is minimal minus full with a paired bootstrap 95% interval over items; discordant counts are items only the full design, or only the minimal one, gets right.

Set	n	Accuracy (%)			95% CI	discordant	p	ECE ↓	
		full	minimal	diff.				full	minimal
Sealed	386	81.3	80.1	−1.3	[−3.6, +0.8]	12 / 7	0.36	0.041	0.060
Easy	48	100.0	100.0	+0.0	[+0.0, +0.0]	0 / 0	1.00	0.003	0.004
Original	72	100.0	95.8	−4.2	[−9.7, +0.0]	3 / 0	0.25	0.099	0.076
Exposed hard	55	58.2	54.5	−3.6	[−12.7, +5.5]	4 / 2	0.69	0.137	0.145
Held-out hard	56	48.2	51.8	+3.6	[−7.1, +14.3]	4 / 6	0.75	0.226	0.217

is +1.8 [−6.5, +9.5] and on exposed hard −2.4 [−10.9, +5.5]. Mean sealed-set ECE is 0.037 against 0.033, and the abstain output still separates unsupported items (area under the ROC curve 0.87 [0.76, 0.95] for seed 0). Removing the listwise head, the ensemble, the bootstrap and both extra loss terms at once changes no set by more than its interval: about ± 2 points on the sealed set and ± 9 on the multi-step sets.

Figure A3 puts the four three-seed configurations side by side.

Does this hold with all the training data? The runs above use 6,000 training items, and the released model was trained on 34,408. A larger head or an ensemble might earn its place only with more data, so we trained the full design and the minimal configuration once each on every training item (Table 5). On the sealed set the full design scores 81.3% and the minimal configuration 80.1%, a difference of −1.3 points [−3.6, +0.8] ($p = 0.36$); on the original tier the full design is ahead by three items of 72 ($p = 0.25$); on held-out hard the minimal configuration is ahead by +3.6 [−7.1, +14.3]. Sealed-set calibration error is 0.041 for the full design and 0.060 for the minimal one. The full-design run is a fresh training run of the released configuration, and it scores 81.3% on the sealed set where the released model scores 82.6%: two runs of one configuration, evaluated on different numerical paths, differ by as much as the two designs do. With one seed per arm and intervals that all contain zero, this neither shows that the full design helps at full scale nor rules out an advantage of one to three points on single-pass decisions; the point estimates on the sealed set, the original tier and calibration lean towards the full design, and settling the question would take several seeds at this scale.

6.4. Summary: what to keep

Our runs support three parts of the design. Slicing the decoder loses no measured accuracy while removing 44% of the layers. Reading options from marker states with a trained head is the mechanism every result here rests on, although we did not compare it with scoring through the language-model head. A dedicated abstain head works only when it is supervised. For single-pass decisions on a backbone of this kind, this gives a short recipe: cut the decoder at the layer where accuracy plateaus, put a small pointwise scorer on per-option states, train an abstain output with its own target, and fit a temperature per question type. We could not detect a benefit from the listwise head, the bootstrap, the ensemble or the extra loss terms, one at a time or all together: the minimal configuration without them is within two points of the full design on the sealed set over three seeds (Table 4). The fact channel is the least settled: its point estimate on exposed multi-step items is negative when removed and inside the noise, and a set several times larger would be needed to decide it.

7. Limits: Where a Single Pass Stops

Two of the requirements are not met everywhere. On questions that need a chain of computation the model is neither accurate nor calibrated, and its answers are not fully invariant to the order of the options.

7.1. Multi-step questions stay near half, whatever we change

Table 1 gives the released model’s results. It is above chance on the multi-step sets and far below its accuracy elsewhere: 48.2% [35.7, 61.0] on held-out hard against 82.6% on the sealed set.

What we tried. Across the 18 controlled runs with at least 18 layers, held-out hard accuracy lies between 44.6% and 62.5%; no configuration reaches two thirds. The released model adds two things to the controlled reference: almost six times more training data, and about 3,800 synthetic items generated specifically for chained date arithmetic, amended policies and multi-hop lookups, with the exposed hard items as structural templates. During development, adding those items moved the sealed set from 79.8% to 82.6% and left held-out hard at 26 of 56 before and after. These are development figures from the GPU fast path, which agrees with Table 1 on the sealed set (319 of 386) and differs by one item on held-out hard. The released model scores 56.4% on exposed hard against 48.2% on held-out hard, a gap in the direction one would expect if the templates helped only on the items they were drawn from, though with about 55 items a side the gap is well inside its noise.

Where it fails. The released model’s errors are concentrated where the question is a calculation chain (Appendix B): over all 111 hard items it answers 6 of 6 adversarial and 5 of 5 routing items but 3 of 15 temporal and numeric items and 4 of 19 long-policy items.

What this does and does not show. We ran no model that is allowed a reasoning chain on these items, so we cannot report from our own runs how much a chain would recover; the benchmark’s public leaderboard lists generative and larger systems well above our figures on its hard tier, which we mention as context and did not reproduce. With 56 held-out items, our runs cannot exclude gains of several points from any single intervention. What they exclude is the outcome we were aiming for: none of depth beyond the plateau, the head, the fact channel, or thousands of targeted synthetic items brought a single-pass model close to its single-pass accuracy on these questions.

7.2. Answers depend on option order for one item in ten

Sensitivity of multiple-choice answers to the order of options is well documented for language models [53], and Li [24] reports that 30% of an open decision model’s decisions change when the options are listed in reverse. We applied the same test to the released model on every choice-type item with a definite answer, re-listing the options in reverse and in one seeded random order (Table A5). On the sealed set, reversing the options changes the predicted option on 9.7% [6.2, 14.8] of 186 items, while accuracy is 83.9% in the original order and 84.9% reversed. A random reordering changes 7.0% [4.1, 11.6]. On the easy and original tiers, where the model is confident, at most two of 36 predictions change under either reordering. The model is therefore not invariant to option order: on the sealed set, about one answer in ten depends on it. The items that change are ones it was unsure of (mean confidence 0.52 in the original order, against 0.90 for the rest), and the changes do not favour either order. The option markers are read causally, so each option state sees only the options before it, and a symmetric treatment of options is a direction we have not explored. Figure A5 shows both robustness checks.

8. Limitations and Future Directions

Limitations. The evidence comes from one backbone family, one language, text only, and one training recipe of 6,000 items and one epoch. Four configurations have three seeds and the other ablations have one, so for those we can bound an effect on the sealed set to a few points but cannot rank small effects, and their intervals carry no seed variance. We ran no baseline that is not our own architecture and no model that reasons step by step. The held-out multi-step set has 56 items and its intervals are about ± 13 points. The sealed set was used to compare development runs, is private, and has only 15 items without a supported answer. The training set is not released, so outside replication of the controlled runs requires other data. All seed-varying runs share one 6,000-item subset, so they measure training noise and not the choice of subset; on the full training set we compared only the full design with the minimal one, one seed each, which cannot detect a difference of one to three points. The size and data-volume observations are single-seed and predate the final data build. Calibration is reported after temperature scaling.

Future directions. Three experiments would settle what this paper leaves open: a baseline that is not our own architecture (the same backbone scoring options through its language-model head, and a generative model that reasons step by step on the multi-step items); several seeds of the full and minimal designs on all the training data; and inference latency measured at each depth. Two design directions follow from the limits: a symmetric treatment of options, to remove the dependence on their order, and a detector for multi-step questions, so that they can be sent to a model that reasons. Beyond these, we plan to pair the single-pass path with a reasoning path inside one model, and to extend the input from text to images and audio.

9. Related Work

Decision models for agents. Jev [2] introduced the class and its interface; its architecture and training method are described only at a high level. Li [24] evaluated Jev and one open-weight model with the same interface across eleven agent decision points. That study reports pooled calibration error of 0.158 for the open model and 0.143 for Jev on six of the decision points (0.198 and 0.080 on the others), and that 30% of the open model’s decisions change when the order of options is reversed. Closely related systems make a single kind of decision: safety classifiers built on a language model

[18, 52], query routers [33], and rerankers that fine-tune a decoder to emit a score [27]. We do not claim a new task: the task and the interface are Jev’s. Our contribution is an open model with a documented design and a controlled account of which of its parts carry weight; Section 7.2 applies the same option-order test to our model.

Intermediate layers and truncation. That deep layers can be removed with little loss is, by now, a repeated finding. For encoders, dropping top layers costs little on classification [37, 40]. For decoders, up to about half the layers can be removed with a short parameter-efficient fine-tune [14, 28]; the last layers can be dropped when the model is used as an embedder [44]; intermediate layers often give the best features [43]; and a classifier on an intermediate layer of a pruned small model beats one on the final layer [42]. Early-exit methods make the depth adaptive [9, 48]. Our slicing result therefore confirms prior work; what we add is a matched, three-seed comparison for multi-option decisions, with calibration reported at both depths and a sweep that locates the plateau. Wang et al. [46] find that layer pruning damages long generated reasoning chains, a different setting from ours; here multi-step accuracy is flat in depth beyond the plateau, but flat at a low level.

Limits of a single forward pass. A fixed-depth transformer without intermediate tokens is limited in the serial computation it can express, and chain-of-thought removes the limit [10, 25, 29]. Empirically, compositional tasks are solved by pattern matching that fails out of distribution [8], the gap between knowing the parts and composing them does not close with scale [38], and models fine-tuned on two synthetic facts fail to compose them, although composition succeeds when one of the facts is natural [3]. A single pass *can* compose k steps when depth grows with k and training is extensive [41, 45, 49, 50]. Our negative result is the applied counterpart: at fixed depth and with thousands of targeted examples, a discriminative model did not learn it, on a held-out split.

Related work on decoders used as classifiers, listwise scoring, calibration, ensembles and abstention continues in Appendix E.

10. Conclusion

We built an open System One decision model from a sliced decoder and trained decision heads, and then measured what each part buys. Just over half the backbone and one pointwise head reach the accuracy of the full design within our intervals; an abstain target is needed for the abstain output; for the rest of what we added we could detect no benefit. On multi-step items nothing we tried came close to single-pass accuracy, and the model’s confidence does not reveal that. We release the model, the code and the per-configuration flags so that these comparisons can be repeated on other data.

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A. Training and Evaluation Details

Controlled recipe. Every controlled run in Section 6 uses Qwen3.5-4B-Base, LoRA rank 16 with $\alpha=32$ and dropout 0.05 on the kept layers, one epoch over a fixed 6,000-item subset of the training set, learning rate 10^{-4} for the adapter and the new token embeddings and 10^{-3} for the heads with 3% warm-up, length-bucketed micro-batches of up to 16,384 padded tokens, bfloat16 autocast and gradient checkpointing. Temperatures are fitted on an 800-item subset of a calibration split that is disjoint from the training set. The 6,000-item subset and the calibration subset are identical across runs; a hash of the subset is stored with each run and was checked. The seed controls adapter and head initialisation, dropout, batch order and bootstrap weights. A run is 291 or 292 optimiser steps, except the run without the fact channel, whose shorter inputs pack into 261 batches, so that ablation also has about 10% fewer updates. Each controlled run took 12 to 33 minutes on one A100 40 GB, and all runs together, including two on the full training set, cost about \$20 of GPU time.

Released model. The released weights use the same architecture at 18 layers with the listwise head, trained on the full training set (34,532 decisions built, 34,408 after the encoder drops items that do not fit the token budget) with temperatures fitted on about 1,800. The adapter is merged, so the release is one 5.4 GB file. All released-model numbers in this paper come from one per-item evaluation of the released file on a consumer GPU without the optimised kernels. The GPU fast path used for the public model card agrees with it on all but a few items (26 against 27 of 56 correct on held-out hard, 58 of 111 on the whole hard tier in both).

Training data. Each item is one decision: a state, one typed question, a target distribution over the options, an abstain flag and sometimes a supporting quote. Roughly two thirds of the items are public datasets with permissive licences recast into this format, keeping annotator disagreement as a soft label where the source records it. Most of the rest are written by code generators that produce a document and compute its answer, so the label is correct by construction (long policies with amendments and effective dates, worked answers with one slip, rosters across daylight-saving changes, and similar). A small remainder is labelled by a blind cascade in which two different language models label every item without seeing each other’s output, a third judges disagreements, and a random tenth of agreed items is audited. The mix is balanced over 15 decision families. The training set is not released.

Leakage control. The sealed set and the 56 held-out hard items were never used for training, for fitting temperatures, or as input to any generator. The sealed set was used to compare development runs, including the choice of which model to release; the held-out hard set was scored during

Table A1 | Depth sweep at the controlled recipe. Accuracy in %, mean over the available seeds; ECE after temperature scaling. At 6 layers held-out hard is below the 33.5% chance level (12 of 56 correct against 19 expected), and exposed hard is at chance at both 6 and 12 layers.

Layers	seeds	Sealed	Easy	Original	Exposed hard	Held-out hard	ECE sealed	ECE held-out
6	1	44.8	79.2	45.8	32.7	21.4	0.062	0.265
12	1	64.5	93.8	73.6	32.7	37.5	0.048	0.156
18	3	79.4	100.0	93.5	52.7	53.0	0.033	0.163
24	1	81.1	100.0	88.9	50.9	53.6	0.033	0.115
32	3	79.4	99.3	92.1	49.7	50.6	0.031	0.177

development as well. Every training item is checked against every evaluation item for 13-gram overlap; 181 candidate items were removed by this check. The 55 exposed hard items are disclosed in Section 4 and reported separately everywhere.

Metrics. Accuracy is top-label accuracy over items that have a definite answer. Calibration error is top-label expected calibration error with ten equal-width bins, after temperature scaling. Intervals on single accuracies are 95% Wilson intervals. For comparisons with several seeds per arm, the interval is a 95% percentile interval from 10,000 bootstrap resamples that draw seeds with replacement within each arm and items with replacement, jointly. For comparisons between two single runs it is a paired bootstrap over items only, and paired tests are exact McNemar tests, uncorrected for multiple comparisons. Intervals on areas under the ROC curve resample items.

B. Depth Sweep and Hard-Tier Families

Table A1 gives the depth sweep of Section 6.1. Table A2 splits the released model’s hard-tier results by family and by exposure; the benchmark’s ten hard-tier families are its own and differ from the 12 sealed-set families and the 15 training families. Families that ask for a judgement are answered well; families that ask for a calculation chain are not, on either half.

Table A2 | Released model on the public hard tier by family: correct over total on the 55 exposed items, the 56 held-out items, and all 111.

Family	Exposed	Held-out	All
Adversarial	3/3	3/3	6/6
Routing (hard)	2/2	3/3	5/5
Trap	4/4	3/4	7/8
Trade-off	2/3	2/3	4/6
Judge (hard)	5/9	6/8	11/17
Ambiguous	2/4	2/3	4/7
Probability	2/4	3/6	5/10
Multi-hop	6/9	3/9	9/18
Long policy	3/10	1/9	4/19
Temporal and numeric	2/7	1/8	3/15

C. Latency Measurement

Latency was measured on a clean install: the public repository and weights at pinned commits, installed into an empty environment on one NVIDIA H100, served over HTTP with the fast path

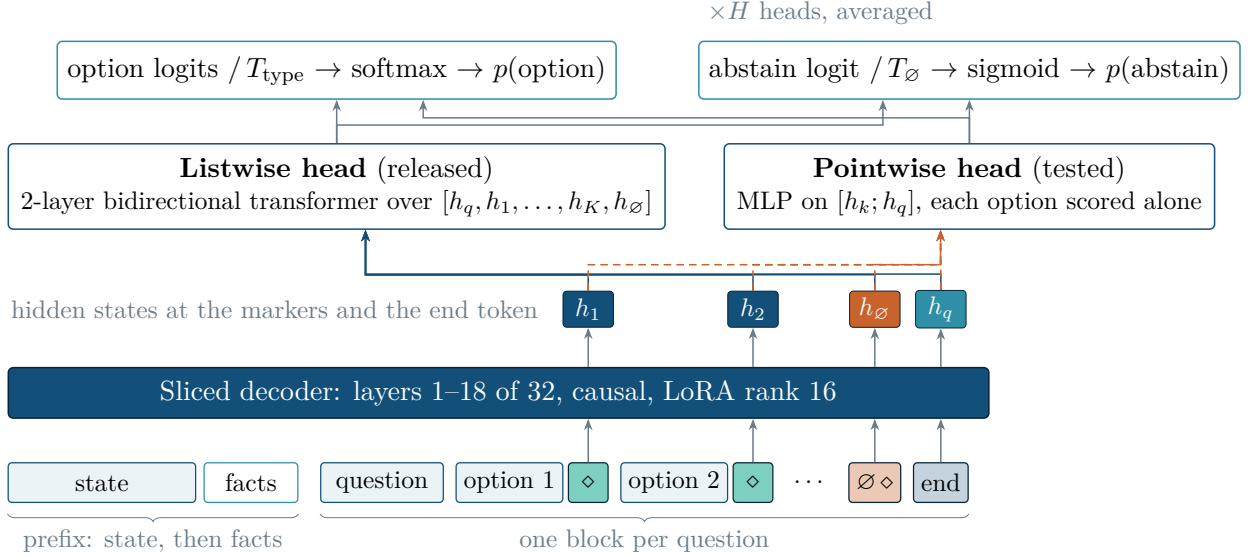


Figure A1 | Input layout and decision head. The prefix holds the state and then its fact lines. In each question block every option is followed by a marker token ($\diamond$); an abstain marker and an end token close the block. The hidden states at the markers and at the end token are the only inputs to the head. The listwise head lets these states attend to each other; the pointwise head scores each option state with the question state alone. H heads share the backbone.

on, and driven by the JevBench client from the same machine, one request at a time. The median over the 231 public items was 15.1 ms end to end (13.4 ms on the easy and original tiers, 29.7 ms on the hard tier, whose states are long) and the 95th percentile was 83.8 ms. Four of the 231 requests took 0.5 to 1 s, all in one length bucket; graphs are captured at start-up, and we have not identified the cause. The public model card quotes about 16 ms from an earlier run of the same kind. On a laptop GPU without the optimised kernels the same model takes about 0.4 s per request.

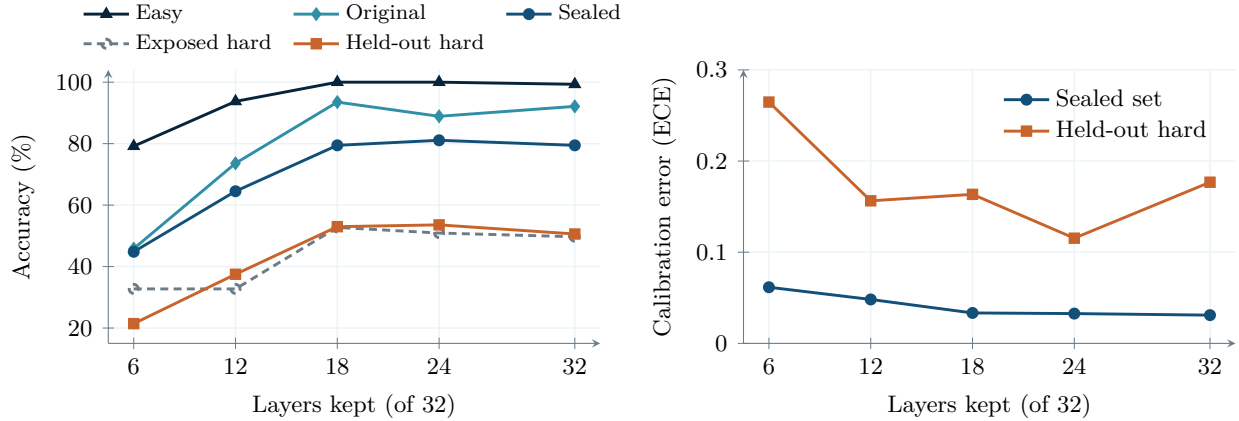
D. Reproducibility

The repository contains the model, trainer, losses, exporter, server and the in-process evaluator used for every accuracy number here. The trainer exposes one flag per ablation: depth (`--cut`), head type (`--head-kind`), number of heads (`--n-heads`), bootstrap (`--no-bootstrap`), fact channel (`--no-facts`), loss weights (`--w-brier`, `--w-emd`, `--w-abstain`), and the two seeds (`--seed`, `--data-seed`). Tables and most numbers in the prose are generated from stored per-item predictions by a script and are not transcribed by hand. The public benchmark tiers are available from their maintainers. The sealed set is private so that it stays usable as a test, and we report it only in aggregate. Because the training set is not released, the controlled runs can be repeated exactly only by us; any dataset in the documented record format trains with the same code.

E. Additional Figures, Tables and Related Work

This appendix holds figures, tables and paragraphs that support the main text.

Decoders as classifiers. Replacing generation by a classification head on a decoder’s hidden states is well established: label-supervised adaptation of a decoder matches or exceeds encoder baselines [26, 51], and decoders become strong text encoders after bidirectional attention is enabled



(a) Accuracy on all five sets against depth (means over seeds). (b) Calibration error against depth, after temperature scaling.

Figure A2 | Depth in more detail. The easy and original tiers saturate by layer 18 as well. Calibration on the sealed set improves with depth up to the plateau and is flat after it; on held-out hard it is poor at every depth.

and a short adaptation is applied [4]. Scoring multiple-choice options through the language-model head is the common alternative, and is known to be sensitive to option order and symbols [39, 53]. Our model sits in the first family and differs in reading one marker state per option, so the number and wording of options are free at inference.

Listwise scoring. Scoring a set of candidates jointly is standard in ranking, from the ListNet loss [5] to groupwise and set-level scorers with cross-candidate attention [1, 36], language-model rankers trained with listwise losses [54], and one-pass listwise rerankers [11]. We adopted the idea for options of a question and then tested it against a pointwise scorer; the pointwise scorer was at least as good (Section 6.2).

Calibration, ensembles and abstention. Temperature scaling is the standard post-hoc fix for overconfidence [15], binned calibration error has known estimation problems [20, 31], and post-hoc calibration does not survive distribution shift [35]. Fine-tuned transformers are typically well calibrated in domain [7]. Deep ensembles [21] and their cheaper relatives that share most parameters [16, 23, 34, 47] are the usual source of an uncertainty estimate. Nixon et al. [32] found that bootstrap resampling gives deep ensembles no advantage over random initialisation alone; our ablation agrees. Abstention has a long line of work as selective prediction [12, 13, 19] and out-of-scope detection [17, 22].

Seed variance is large on small sets. On held-out hard the 18-layer seeds span 51.8–53.6% and the 32-layer seeds span 44.6–58.9%; on exposed hard the ranges are 47.3–56.4% and 38.2–56.4%; on the original tier it is the 18-layer runs that vary more (90.3–98.6% against 88.9–95.8%). Three seeds do not support a claim about which depth is more stable. They do show that a single-seed depth comparison on these sets can come out several points either way, which is how we first read the question.

Size and data volume. Two earlier single-seed runs, on a previous build of the training data and so not comparable in level with Table 2, found a 9B backbone of the same family cut at half depth no better than the 4B one on the sealed set (76.2% against 79.5% for the 4B run on that build) and all of that earlier build, about 32,000 items, no better than 6,000 of them (79.8% against 79.5%). These are observations from one seed each and we draw no conclusion from them.

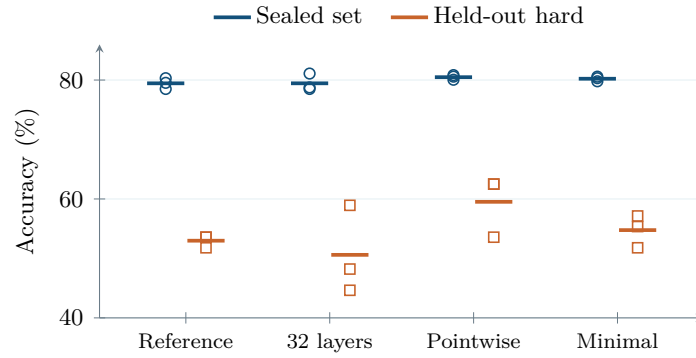
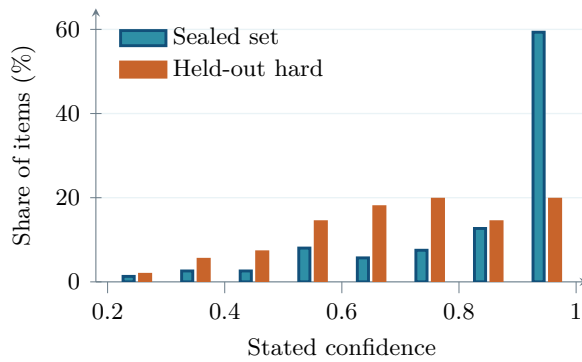
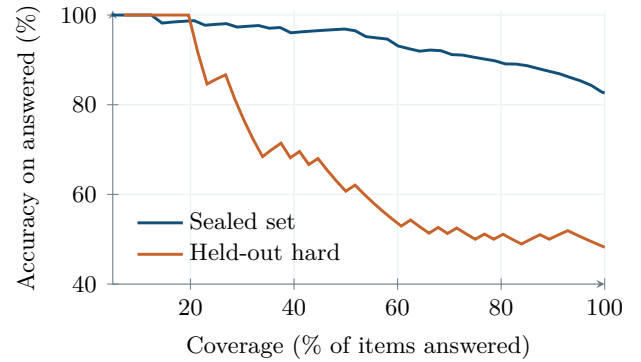


Figure A3 | The four configurations that have three seeds each. Horizontal bars are means and open markers are the seeds. On the sealed set all twelve runs lie between 78.5% and 81.1%; on the 56 held-out multi-step items seeds of one configuration differ by up to 14 points. Intervals for the differences are in Tables 2, A3 and 4.



(a) Distribution of stated confidence.



(b) Accuracy when only the most confident items are answered.

Figure A4 | Confidence on single-pass and multi-step items. On the sealed set 59% of answers carry confidence above 0.9; on held-out hard the confidence is spread between 0.5 and 0.9, where most of its errors lie. Thresholding raises sealed-set accuracy steadily and helps the multi-step set only at low coverage.

Table A3 | Pointwise against listwise head, three seeds each at 18 layers. The difference is pointwise minus listwise; its interval resamples seeds and items.

Set	n	Listwise		Pointwise		diff.	95% CI
		seeds	mean	seeds	mean		
Sealed	386	79.5 / 78.5 / 80.3	79.4	80.6 / 80.8 / 80.1	80.5	+1.0	[−1.1, +3.2]
Easy	48	100.0 / 100.0 / 100.0	100.0	100.0 / 100.0 / 100.0	100.0	+0.0	[+0.0, +0.0]
Original	72	90.3 / 98.6 / 91.7	93.5	91.7 / 94.4 / 87.5	91.2	−2.3	[−9.7, +3.7]
Exposed hard	55	54.5 / 56.4 / 47.3	52.7	60.0 / 52.7 / 49.1	53.9	+1.2	[−8.5, +10.9]
Held-out hard	56	51.8 / 53.6 / 53.6	53.0	62.5 / 62.5 / 53.6	59.5	+6.5	[−3.0, +16.1]

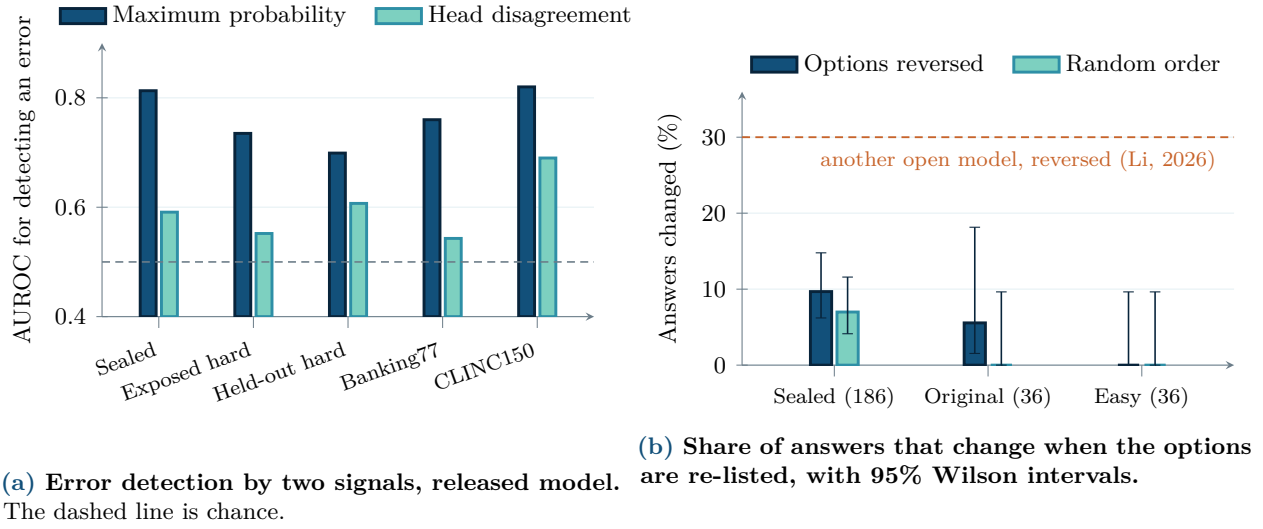


Figure A5 | Two robustness checks on the released model. Head disagreement is a weaker error signal than the maximum probability on every set. Re-listing the options changes about one sealed-set answer in ten and almost none on the easier tiers.

Table A4 | Area under the ROC curve, with bootstrap 95% intervals, for detecting an error (first four columns) and for detecting an item with no supported answer (last column; 15 such items). 0.5 is chance.

Variant	Sealed: error		Held-out hard: error		Sealed: abstain
	max. prob.	disagr.	max. prob.	disagr.	abstain prob.
Bootstrap ensemble (reference)	0.82 [0.77, 0.86]	0.63 [0.56, 0.70]	0.80 [0.67, 0.91]	0.56 [0.40, 0.71]	0.82 [0.70, 0.93]
Ensemble without bootstrap	0.82 [0.77, 0.86]	0.66 [0.59, 0.72]	0.70 [0.55, 0.83]	0.60 [0.46, 0.75]	0.85 [0.75, 0.93]
Single head	0.81 [0.76, 0.86]	–	0.79 [0.66, 0.90]	–	0.88 [0.81, 0.94]
No abstain supervision	0.82 [0.77, 0.87]	0.67 [0.60, 0.74]	0.78 [0.65, 0.89]	0.51 [0.36, 0.67]	0.46 [0.32, 0.60]

Table A5 | Option-order test on the released model, choice-type items. “Changed” is the share of items whose predicted option differs from the original order, with a 95% Wilson interval.

Set	n	Original	Reversed		Random order	
		acc.	changed ↓	acc.	changed ↓	acc.
Sealed	186	83.9	9.7 [6.2, 14.8]	84.9	7.0 [4.1, 11.6]	84.9
Original	36	97.2	5.6 [1.5, 18.1]	97.2	0.0 [0.0, 9.6]	97.2
Easy	36	100.0	0.0 [0.0, 9.6]	100.0	0.0 [0.0, 9.6]	100.0

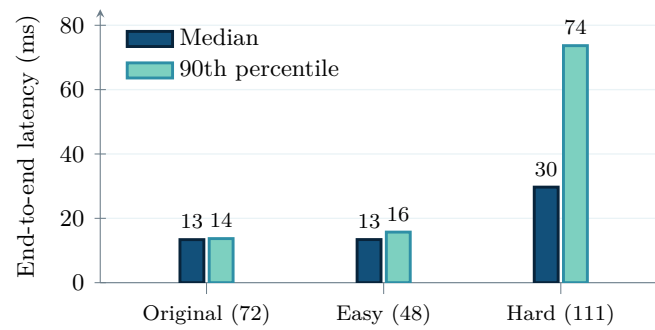


Figure A6 | End-to-end latency of the released model by benchmark tier, on one H100 over HTTP, one request at a time. Hard-tier states are long. The 90th percentile is shown because four slow requests (0.5 to 1 s, all in one length bucket) dominate higher percentiles within a tier.